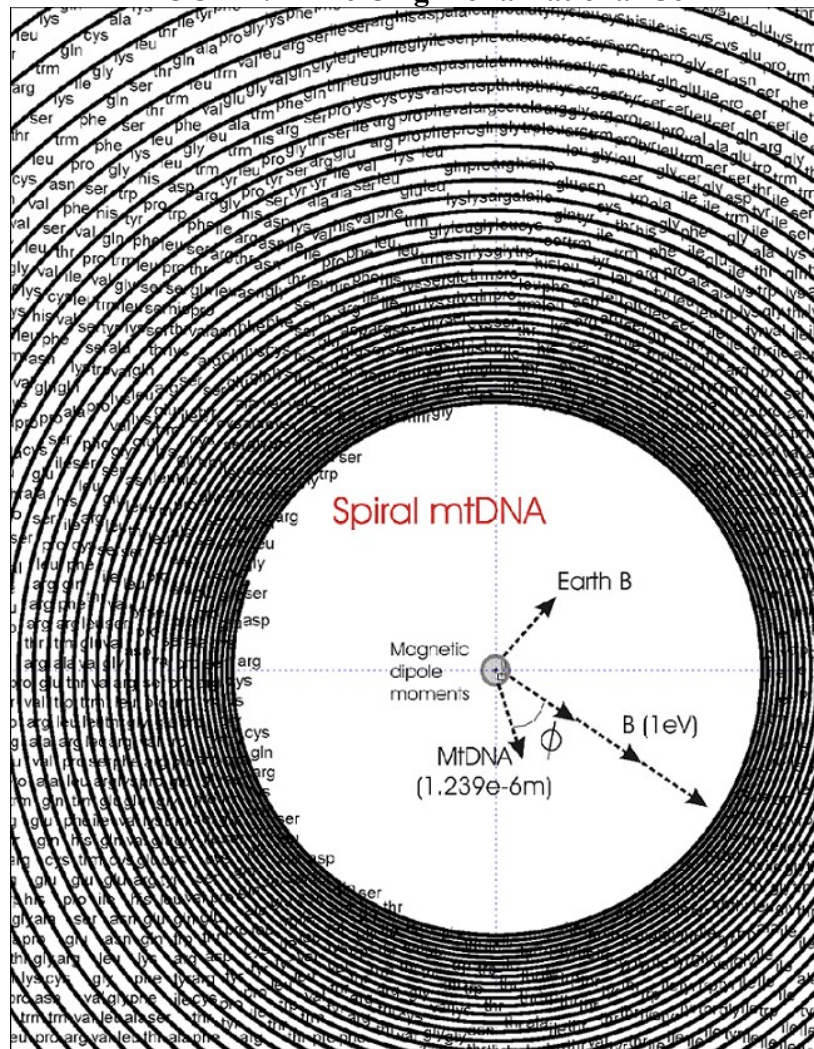


VOL I: “Cell Algebra”

BOOK 1: “The Origin of a Rational Cell”



By Abdon EC Bishop

Quote “When our universe first used the miracle of arithmetic, the construction power of algebra was utilized to weave into the early fabric its intrinsic subfields of ideal prime patterns“

Quote “The initial event of near triple point H₂O state and property of superconductivity is the miracle of perpetual life and the theory of cell evolution”

Quote “Pi is infinite factorization of solid 3-space factoring prime points that generates smooth manifolds in transfinite polynomial time less than four seconds”

Quote “The evolving universe, think the infinite prime product equal to the first transfinite prime relatively prime natural number in a continuous state of infinite factorability or factorization”

Original Work/Article Submission

Continuing the protected copyright registration number "1041116"
 Dated 07/09/2006
 Titled "Computational Biology: The Biophysics of Bioinformatics"
 Submitted by Christopher Eric Abdon Bishop, Datagrate Solutions

Patent Office Contact:

I wish to register the scientific question with a private DGS funded answer to the Canadian Intellectual Rights Patent/Copyright Office.

"Is Nature a characteristic zero field expressed as a group of relatively prime subfield extensions that find model in the mtDNA of a mitochondrion?"

And forward **Yes**, answering why in an article conditionally Titled

"Only if perpetual Mother Earth's nature is a mobius connected mtDNA superconductivity space that centers a geodetic timeline of magnetically switching groups of haploid chromosomes superconducting at the 'Origin of Life' point"

Submitted by

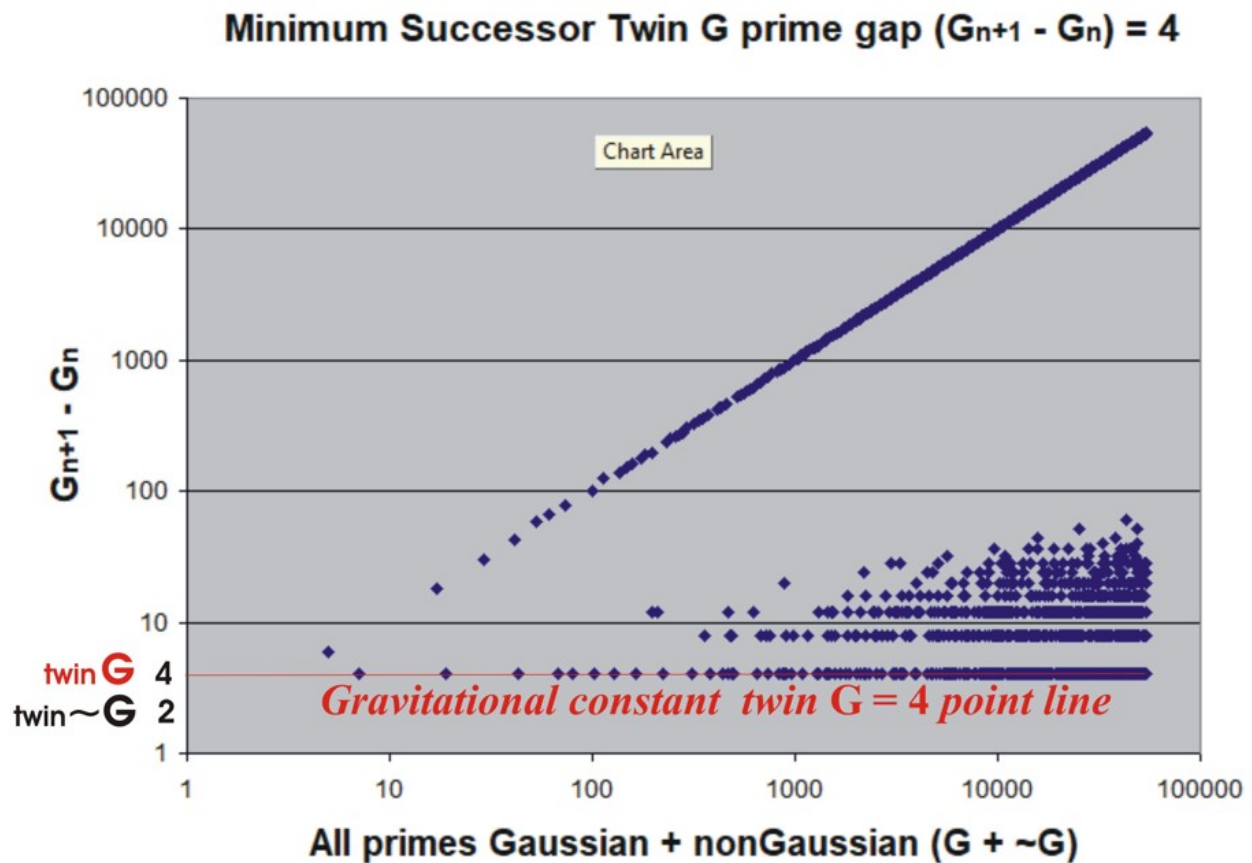
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Personal Quote: *"Natures arithmetic model is a characteristic zero infinite field embedding a submodel of finite incomplete p-groups of relatively prime subfield extensions"*

CELL ALGEBRA MAPS EXONS CUTTING CHROMOSOME

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Abstract

Modern computers model programmable calculators performing arithmetic with a characteristic two algebra that is a vector space over a finite field with $|\Phi(1)|$ primitive roots¹ modulus two. Arithmetic is an abstract algebra only when its modulus has the characteristic of a prime number or zero. Generalizing, there exist higher prime k characteristic programmable calculators performing arithmetic in a vector spaces with $|\Phi(k-1)|$ primitive roots. There is only one primitive root of a characteristic three algebra. The primitive root 2 is the only cyclic generator of the group $\Phi(3)$ with order $|\Phi(3)| = 2$, consequentially the union of $2^1 \bmod(3) = 2$ and $2^2 \bmod(3) = 1$ equals the group $\{1, 2\} = \Phi(3)$.

Introduction

Mathematically a cell is modeled as a programmable calculator that utilizes a characteristic three algebra to perform quantum calculations computed in powers of the primitive root $|\{\text{neutron (dud)}\} \equiv 0, \text{proton (duu)} \equiv +1\}|$ spanning a three dimensional vector space.

The rotator range of Legendre function $(\frac{n}{k}) \bmod(k) \in \{-1, 0, 1\}$ maps figure ~0's Turing Register $3'$ exon streams natural numbers $n = n + 1 \in \mathbb{N}_0$, into three Legendre function output rotation angles. The Legendre function range is interpreted to be a radiant measure of Euclidean arc rotation angle equal $i(\frac{n}{k})$ or $-1, 0, +1$ radians of π translation. The Legendre function rotates n when $(\frac{n}{k}) \bmod(k) = \pm 1$. Only when k/n , is $(\frac{n}{k}) \bmod(k) = \zeta(s) = 0$ selected. Riemann conjectured that $(\frac{n}{k}) \bmod(k) = \zeta(s) = 0$ is satisfied only when $s = \frac{1}{2} + i(\frac{n}{k}) = \frac{1}{2} + i \cdot 0$ and all prime are coincident with the point $(\frac{1}{2}, 0)$. Consequentially if a successor odd natural number $n = n+2$ tests prime, the prime number theorem states that the prime number can be added to a list of $(n/\log(n))$ numbers between 1 and n that tested positive for prime. A result in number theory states that three is a quadric residual for odd primes of the form $12 \cdot n - 1$ and $12 \cdot n + 11$.

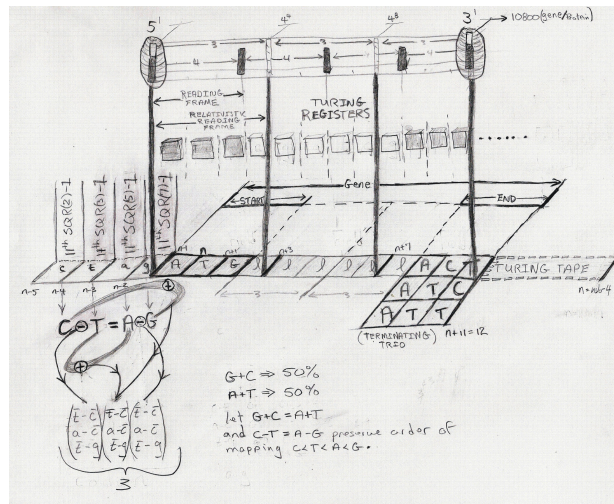


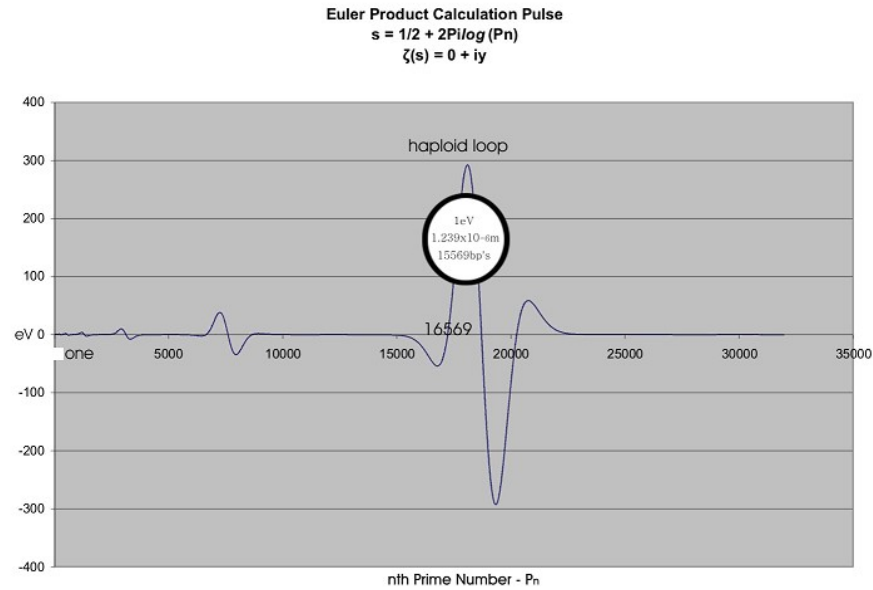
Figure ~0'

A protein is a ternary coded polynomial/Euler polyhedron constructed in a characteristic three cell algebra that can be interpreted as composite prime $p = e^{(\frac{1}{2} + i(\frac{n}{k})) \log 10} = 12 \cdot n - 1$ or $12 \cdot n + 11 = p$ factorized by both a quadratic residual and 'p's cyclically generated primitive roots, both factorized by three dividing a chromosome into genes and exons capable of reducing the protein into primitive prime ideals.²

¹ $\Phi(n)$ is Euler's phi function determines the group of natural numbers that do not factor n . When n prime, the order of the group $|\Phi(n)|$ is maximal equal $n-1$.

² All algebras divide infinity into a composition of rank one finite cyclic p-group only when p prime.

Algebra



Algebra is a vector space over a field of points. The problem is to develop a mapping between a polynomial $P(a)$ and protein $p(a)$ by defining a protein to be the analytical continuation of a gene. How is this accomplished? First we have to recognize that to analytically continue a field F is to extend the field to E . The derivative of an analytically continued field extension E (*protein*) is a subfield F (*gene*), conversely the integral of a field F (*gene*) is an extension field E (*protein*).

A gene is a polynomial $P(F)$ defining a vector subspace whose dimension is determined by the cardinality of the basis set that spans the subspace. The vector subspace is over a scalar field F that has both arithmetic characteristic 'zero' $\equiv \text{mod}(1/0)$ or arithmetic characteristic 'prime' $\equiv \text{mod}(P_n)$ rational boundary points. The fundamental theorem of algebra states the polynomial $P(a) = 0$ equals an algebraic field extension number $E(P(a))$. This means there is a set of complex roots $\{a_1, a_2, a_3, \dots, a_n\}$ that satisfy $P(x) = 0 = (a_1 - x)(a_2 - x)(a_3 - x) \dots (a_n - x)$. While satisfying both the polynomial $P(x) = b_0 \cdot (x - a_0)^0 + b_1 \cdot (x - a_1)^1 + b_2 \cdot (x - a_2)^2 + \dots + b_{n-1} \cdot (x - a_{n-1})^{n-1} = -1$ and the Euler function $\Phi(n) = n - 1$, generates a solution field with negative polarity at measure $x = 1.629 \cdot 10^{-6}$ meters closed loop mapping the normalized diameter of a human mitochondrion mtDNA torus shape. When selecting only those n for which $n = p$ is true or satisfied for a polynomial field extension $(E(P(a))) \text{mod}(4) = 3$ Gaussian prime endpoint orbiting a 16569th cyclic prime degree term that maps the number of base pairs of closed single strand haploid mtDNA molecule.

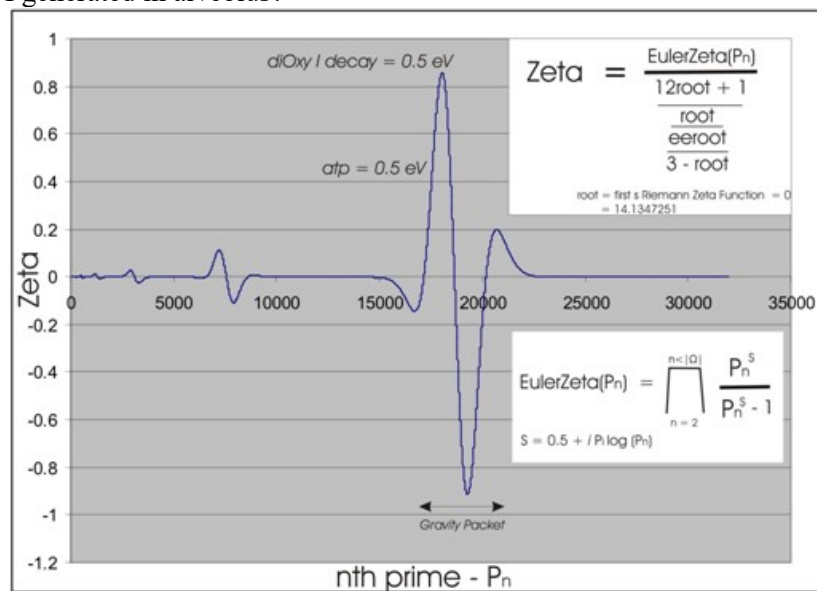
The fundamental theorem of algebra in an extension field of dimension maximally determined by the number of solution roots of 16569th polynomial field $P(F)$ determined by polynomial greatest exponent degree $n = n - 1 = \Phi(n) = \zeta(s) - 1$ only when $s = 1/2 + i \cdot y \cdot \log(P_n)$ inputs $\zeta(s)$ and $\zeta(s) = 0$ are true while $n = p$ a prime and $1/2 + i \cdot y \cdot \log(P_n)$ points prime p is correct.

A point is either a ring set of even-not-prime and odd-not-prime elements that is also a field element only when the characteristic of ring is prime. The point is a ring element that is not a field element when the characteristic of ring is composite or not prime. A vector is a diploid pair of n -tuple numbered endpoints. Each endpoint haploid n -tuple is a point that is a ring point. By default for all vectors one n -tuple point there is always a point that is a ring point and a field point that can be put into coincident with zero point. The other n -tuple endpoint can alternate either as a ring point that is a field point, ie; even-not-prime, or a ring point that is not a field point ie; odd-not-prime.

A vector is always determined by a pair of endpoints. Always one endpoint is both a ring point and coincident field point both with characteristic zero. The other endpoint is a choice point that has characteristic prime only when the point is a ring point that is also a field point, otherwise the chosen point is a ring point that is not a field point and has characteristic not-prime.

An algebras characteristic operates on vectors with two paired endpoints that have two positive integer number characteristics. One endpoint has a constant characteristic zero, the other endpoint a variable characteristic that is a natural number that is either prime or not prime composite. The set of all vectors all intersect a pair of endpoints, one the characteristic zero endpoint, the other an omega or less numbered set of vectors. The first endpoint locates where no intersection ie; the empty set occurs. The second endpoint intersects in a ring. A vector is either open when the non-zero endpoint has characteristic not-prime composite ie; electric field point, or closed vector when the non-zero endpoint has characteristic prime ie; electromagnetic field point.

An analytical geometry with critical point in the alveolus of the respiratory membrane of all mammals exists where virtual molecules of diOxygen I are biologically manufactured. A set of field points is hypothesized to biologically exist that maps the ratio of stem cells to respiratory membrane diOxygen I molecules each point projecting vector tangents with slope determined by the sound of rhymic tones of animal pulmonary ventilation cycle. In refutive feedback, a mammalian respiratory question can be asked. Is the product of applied body stress, and body regenerative capacity, factored by the moles of virtual diOxygen I generated in alveolus?



Cell Algebra

A numeric cell model based upon the arithmetic of natural numbers and the theory of quadratic residuals is utilized to provide an algebraic structural framework capable of consistently mapping exons intersecting a chromosome. A characteristic two modulation maximally intersects the complex plane in two abelian ring points that extends to a vector space over a super-ring of field points divisible by some exponential s extending cyclic powers of a characteristic two subfield. All primes greater than four covered by a Boolean algebra are members of a communicative division ring with unity having no unit member nilpotent $0 \neq k^n$, and all unit members idempotent $k = k^2$. The elements of the Boolean algebra can be generalized to members 0 and ~ 0 labeling endpoints for an infinite set of prime unit intervals all vectors. When $\sim 0 = 1$ the algebra maps a Boolean characteristic two subfield. The intersections of natural numbers n and $k \in \{1, 2, 3 \dots\} = \mathbb{N}_0$ computes the communicative product $n \bullet k = k \bullet n = 1/\sim 0 \in \mathbb{N}_0$ having all near zero neighborhood product solutions determined by the inductive proof generating function $n = n + 1$. The Boolean algebra is modeled by a complex arithmetic operating as a binary coded decimal Gaussian arithmetic calculator capable of adding, subtracting, multiplying and dividing vectors as units of complex numbers in Argand plane.

The calculator can be designed compute finite field solutions solving the Euler zeta function sum and product computations. The calculator output captured reduces Riemann zeta function to the Euler zeta product. The rate of zeta function convergence is fastest for the Euler product computation and slowest

for Euler sum computation. Both function converge have different rate of convergence congruent to the same complex point after $\omega = |\Omega|$ terms. Both functions input the same complex number $s = (s \cdot x) + (s \cdot y) = x + i \cdot y = \frac{1}{2} + i \cdot \pi \cdot \log(P_n)$.

The following C++ code object computes the Euler zeta product and sum functions.

```

class complex{
public:
    double sx ;
    double sy ;
    double re, im;
    complex(double r, double i) { re = r; im = i; sx = 0.5; sy =
54.373006820492399595974564663248;} //147.80115639882421754701165059976;}
//22.459157718361045473427152204544;} //2.0943951023931954923084289221863;} //
6.283185307179586476925286766559 * 2;} //14.523298206623843725990092854029;} //
2.3114546995818434358201863666254;} // 17.079468445347134130927101739093 ;} //
6.283185307179586476925286766559;}
    complex(double r){ re = r; im = 0;}
    complex(){ re = im = 0;}
    friend complex operator+(complex,complex);
    friend complex operator*(complex,complex);
    friend complex operator-(complex,complex);
    friend complex operator/(complex,complex);
    complex Integer(int n);
    complex EulerZeta(complex);
    complex ZetaProduct(complex);
private:
    static complex IsPrime();
    static complex ExtZetaEuler();
};

complex operator+(complex c1,complex c2){
    return complex(c1.re+c2.re,c1.im+c2.im);
}
complex operator*(complex c1,complex c2){
    return complex(c1.re*c2.re-c1.im*c2.im , c1.re * c2.im + c1.im*c2.re);
}
complex operator-(complex c1,complex c2){
    return complex(c1.re-c2.re,c1.im-c2.im);
};

complex operator/(complex c1,complex c2){
    return complex( ((c1.re * c2.re) + (c1.im * c2.im))/(((c2.re * c2.re) + (c2.im * c2.im))) ,
        (((c1.im * c2.re) - (c1.re * c2.im)) / ( (c2.re * c2.re) + (c2.im * c2.im) )) );
};

complex main(complex c)
{
//Omega SUM (1/nEXP(s)) = SUM(nEXP(-s)) = SUM((nEXP(-x)) * (Cos(-ylogn) + i Sin(ylogn)))
//Omega PRODUCT (PnEXP(s)/(PnEXP(s)-1)) 1.072877 5.783639
// 0.99 exp 1000000 = 1.5652998726725347093812599749531e-4365
// 1.01 exp 1000000 = 2.3647358888701483369966440045217e+4321
//log(2.3647358888701483369966440045217e+4321) = 9950.3308531680828482153575442607
//9.2053610811744987926638176113024 = log(9950.3308531680828482153575442607)
//log(9.2053610811744987926638176113024) = 2.2197860405506233931619233570839 > 1
// log(2.2197860405506233931619233570839) = 0.79741081310916781721532214653588 < 1

```



```

FILE *VaryIm;
VaryIm = fopen("VaryIm.csv", "w");
complex EulerProduct(1,0);
complex retProd,retSum;

std::cout << "(" << EulerProduct.ZetaProduct(EulerProduct).re << ","
<< EulerProduct.ZetaProduct(EulerProduct).im << ")";
std::cout << "\n";
retProd = EulerProduct.ZetaProduct(EulerProduct);
fprintf(VaryIm, "%f, %f\n", retProd.re , retProd.im);
fclose( VaryIm );
complex Sum(0,0);
retSum = Sum.EulerZeta(Sum);
return retSum; }

```

The Euler zeta product graph is a superGroup of finite field simple groups each having prime order P_n . All simple field groups satisfy the mathematical definition of a ring with characteristic prime extension fields. Inputting $s = \frac{1}{2} + i\pi \log(P_n)$ the Euler product outputs the following three graphs. Isolate on the graph the n^{th} prime number 15569 requires three AMP per prime number. The decay ATP to ADP to AMP releases $\frac{1}{2}$ eV per transition event. The total haploid base pairs computes three haploid bases times two diploid equal six DNA molecules of D(AMP)A in two haploid subgroups D(AMP)A and D(TMP)A. To generate ATP, molecules of diOxygen I are preferentially selected by mitochondrion over diOxygen III. Once selected the 1eV photon geodesic maps AMP + $\frac{1}{2}$ eV to ADP and ADP+ $\frac{1}{2}$ eV to ATP.

SEE FIGURES next pages

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